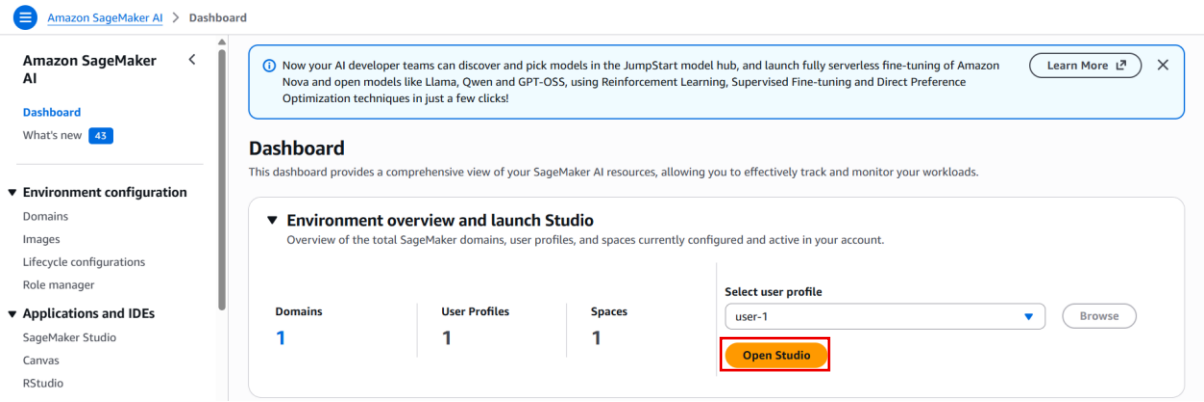
SageMaker Canvas Using AutoML

To Begin with the Lab

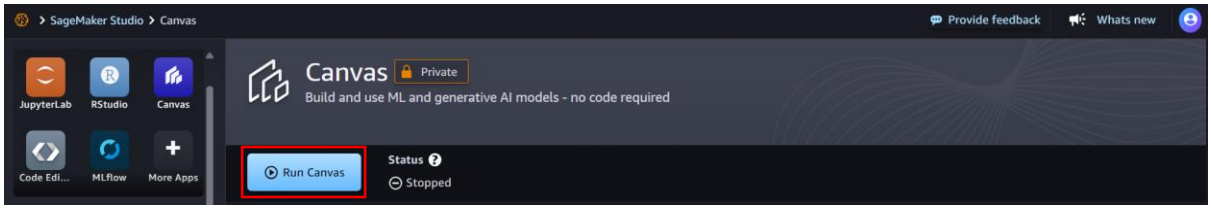
Summary of the Lab

In this lab, you used Amazon Web Services Amazon SageMaker Canvas to build and deploy an AutoML predictive analytics model for income prediction. You configured datasets, target columns, RMSE metrics, and Quick Build training, then analyzed feature importance, explainability visualizations, and advanced metrics to evaluate model performance. Finally, you performed batch and single predictions, deployed the model using SageMaker endpoints, and configured scalable inference instances for real-time application integration and predictions.

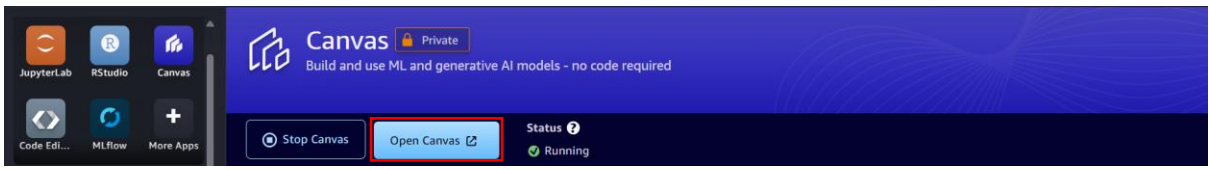
- Log in to AWS Console.
- Open Amazon SageMaker.
- Launch your user profile.



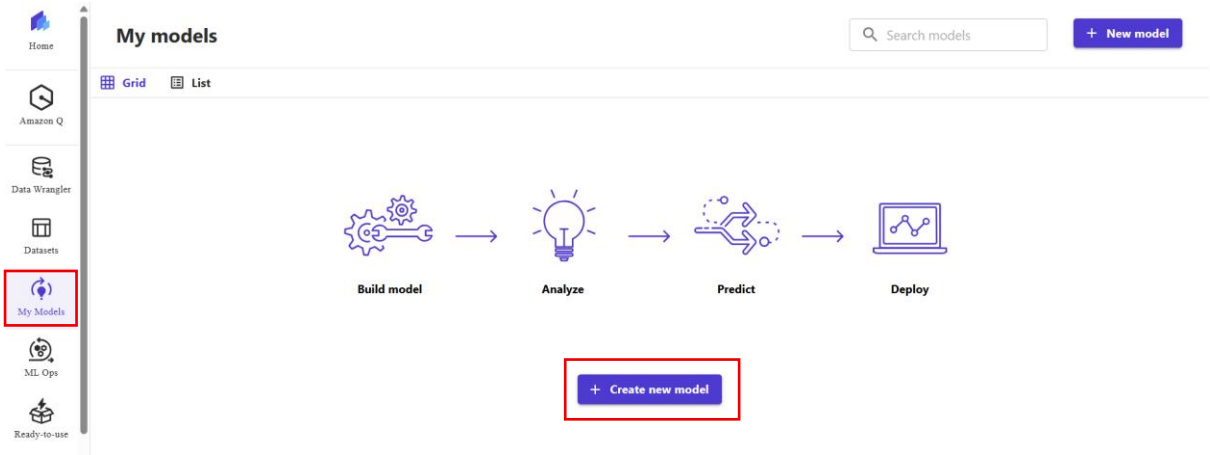
- From Studio applications, locate Canvas.
- If status is Stopped, click Start.



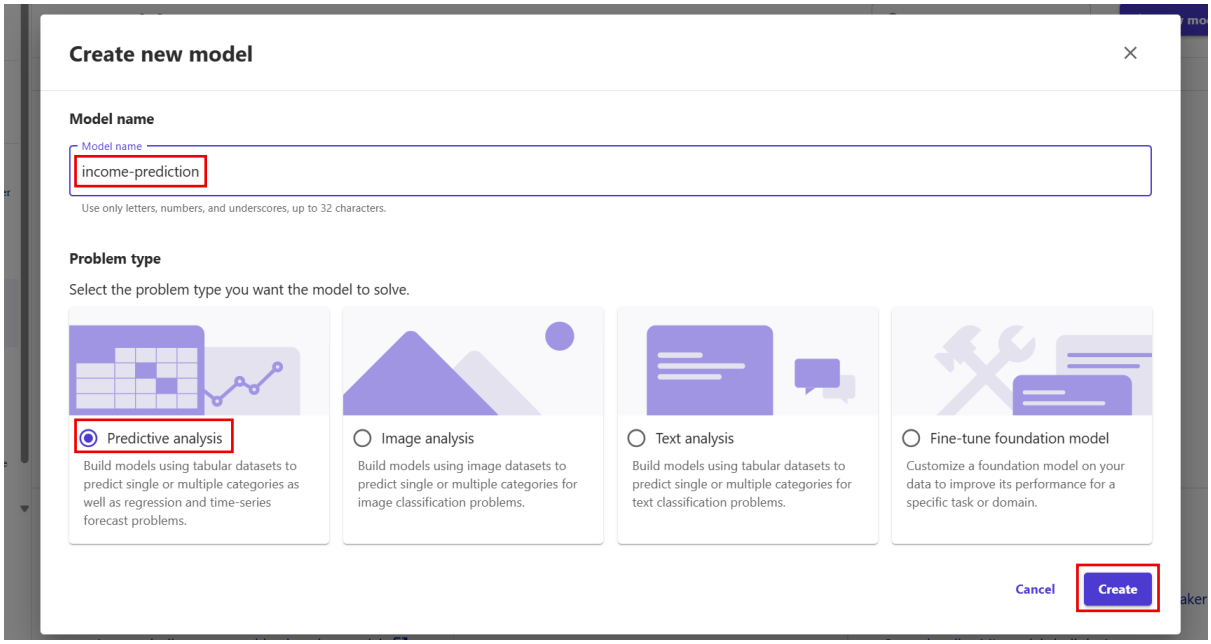
- Wait until status changes to Running.
- Click Open Canvas.



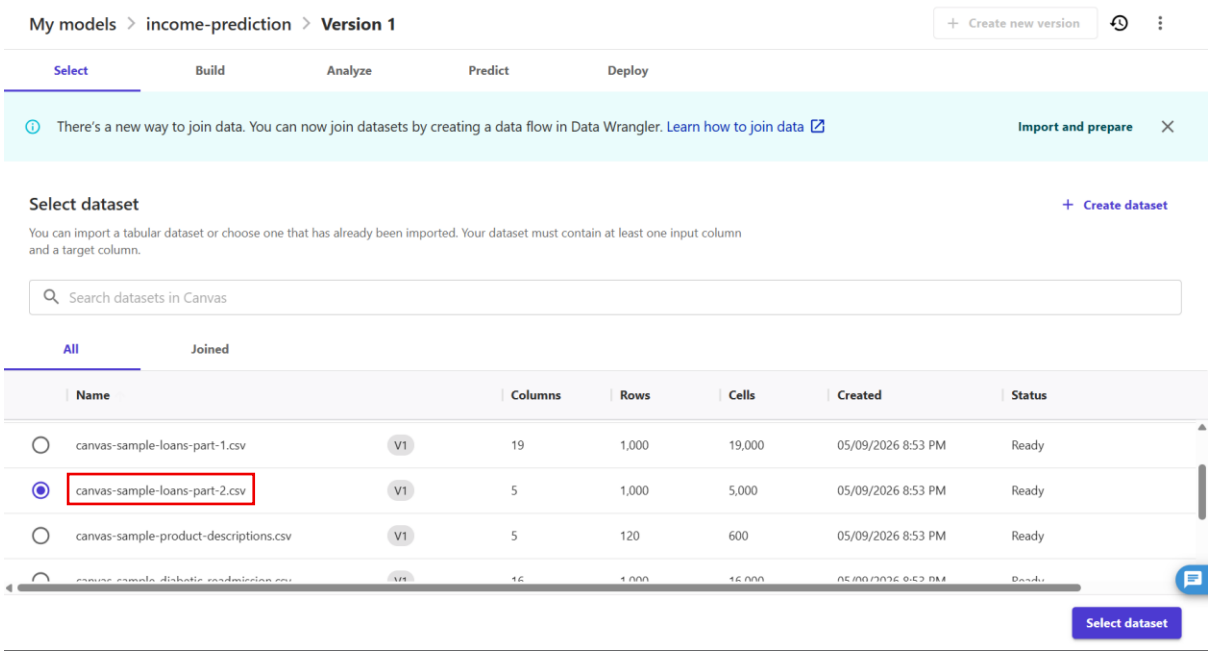
- Inside Canvas, use the left navigation panel.
- Go to My Models



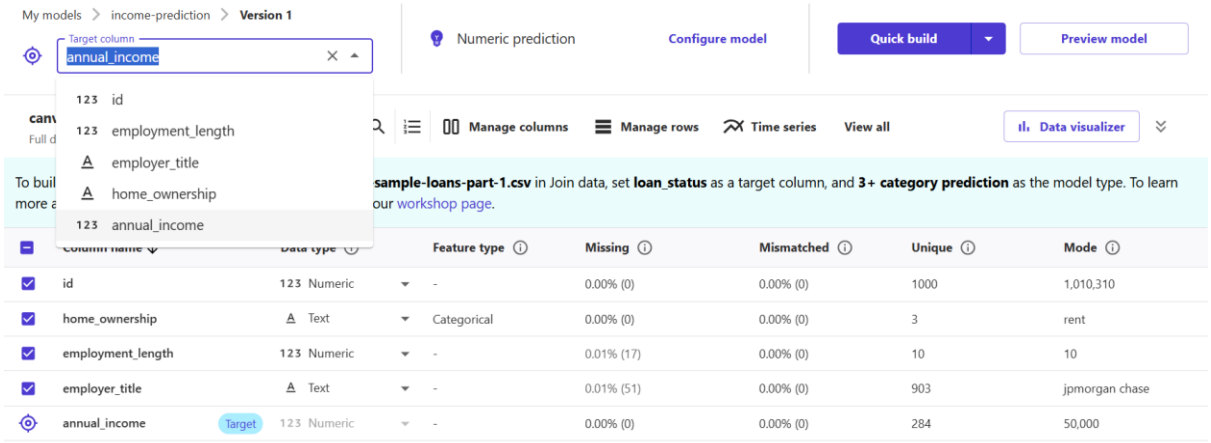
- Click New Model
- Enter model name:
- income-prediction
- Choose:
- Predictive Analytics
- Click Create



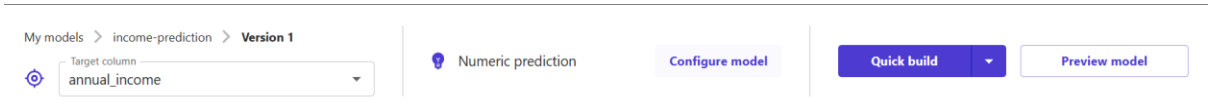
- Choose dataset:
- Canvas Sample Loans Part 2
- Preview the dataset.
- Click:
- Select Dataset
- Next



- Configure the Prediction Model
- Target Column
- Select:
- Annual Income
- This is the value the model will predict.

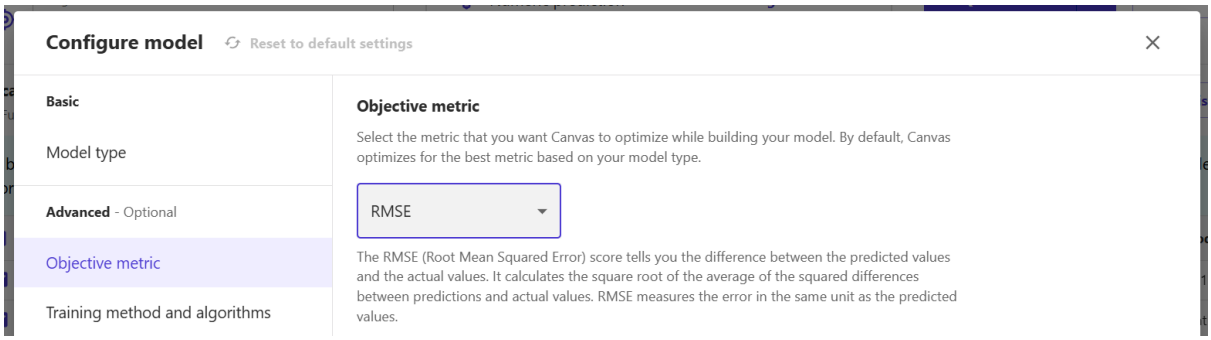


- Configure Model Settings
- Click Configure Model to customize settings.
- Recommended Settings
- Model Type

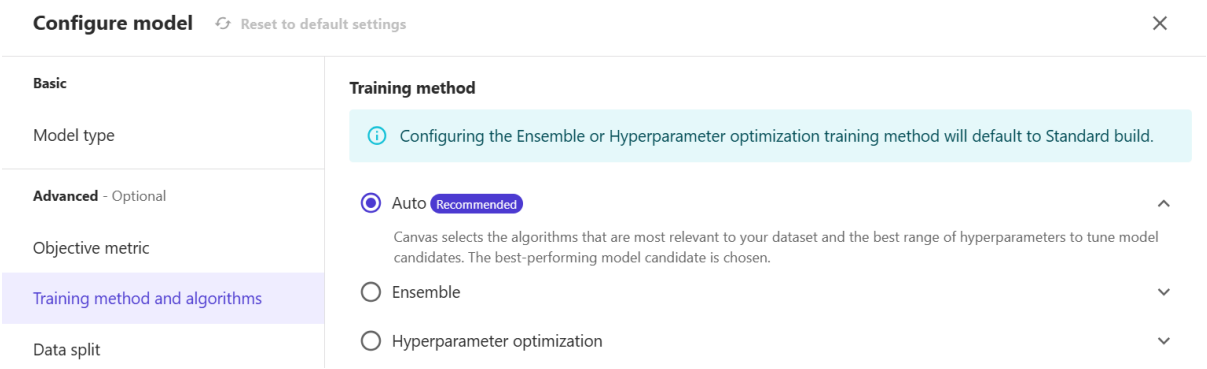


- Numerical prediction (recommended automatically)
- Objective Metric

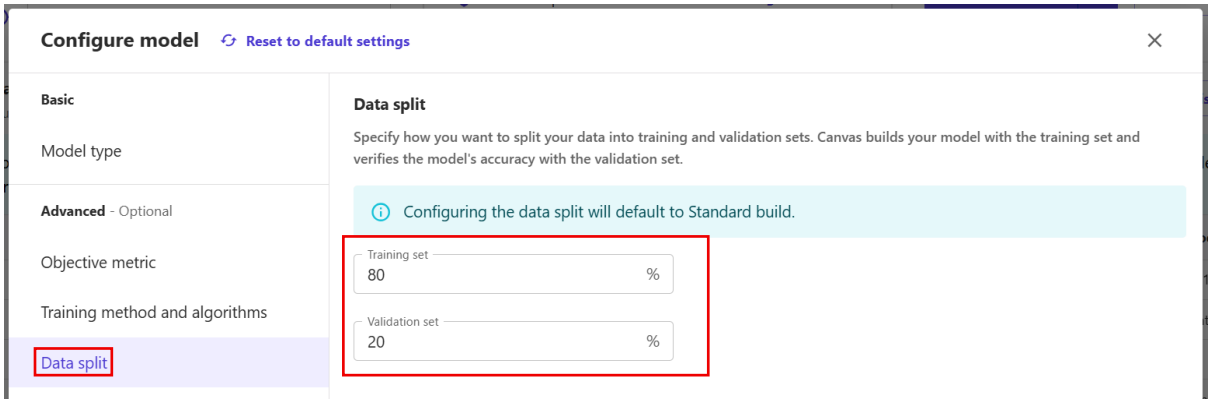
- Choose:
- Root Mean Squared Error (RMSE)



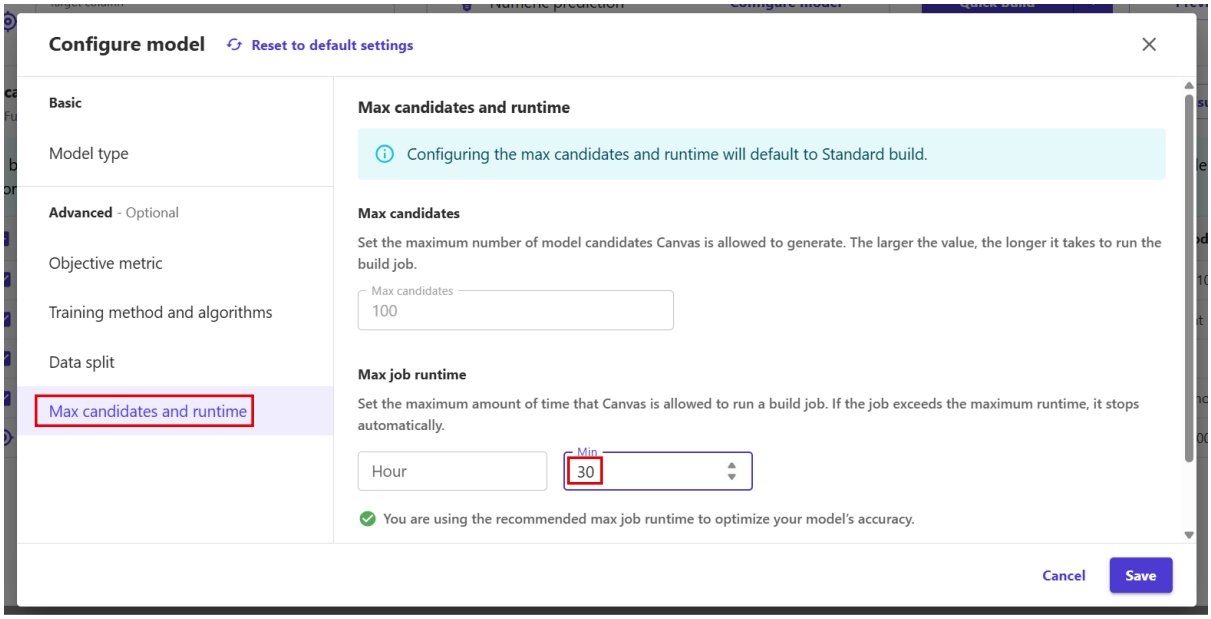
- Training Method



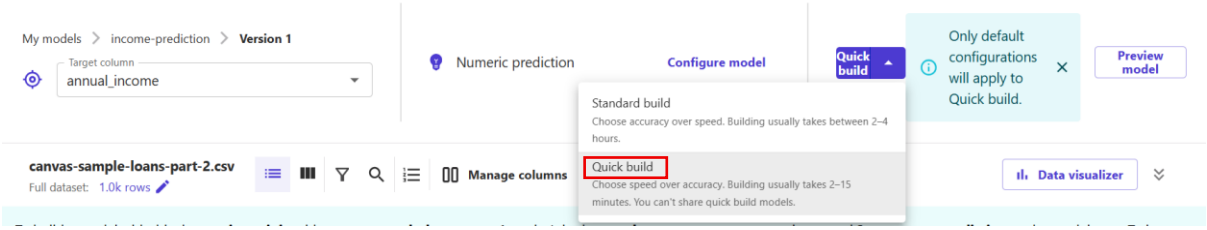
- Default settings are acceptable:
- Training Data- 80%
- Validation Data- 20%



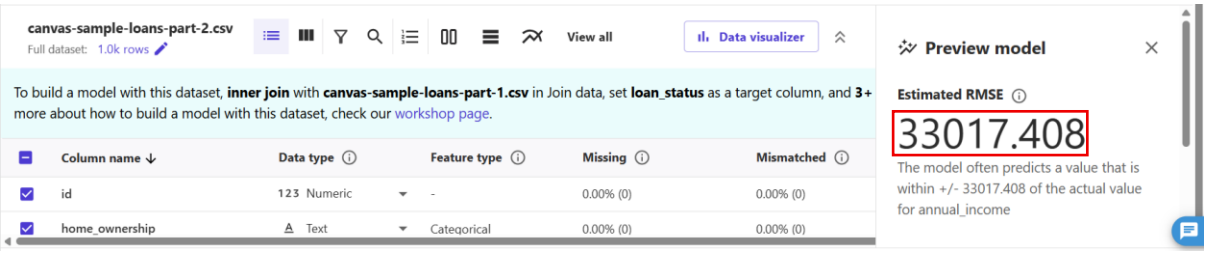
- Configure Runtime
- Optional:
- Set maximum runtime (example: 30 minutes)



- For this lab:
- Keep default settings.
- Use Quick Build

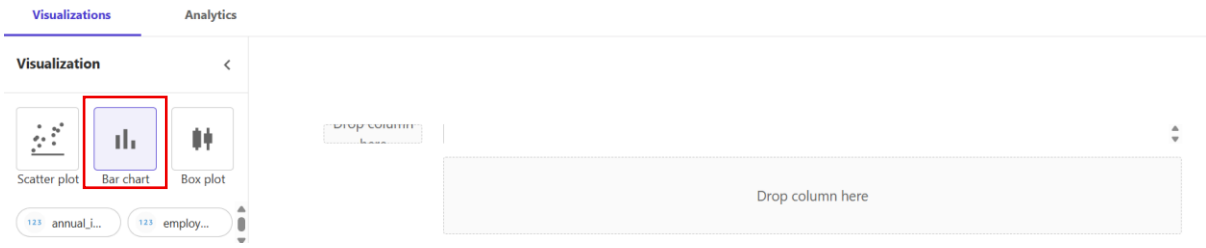


- Before building, click:
- Preview Model
- Canvas quickly estimates performance.
- Example output:
- Estimated RMSE ≈ 33,000

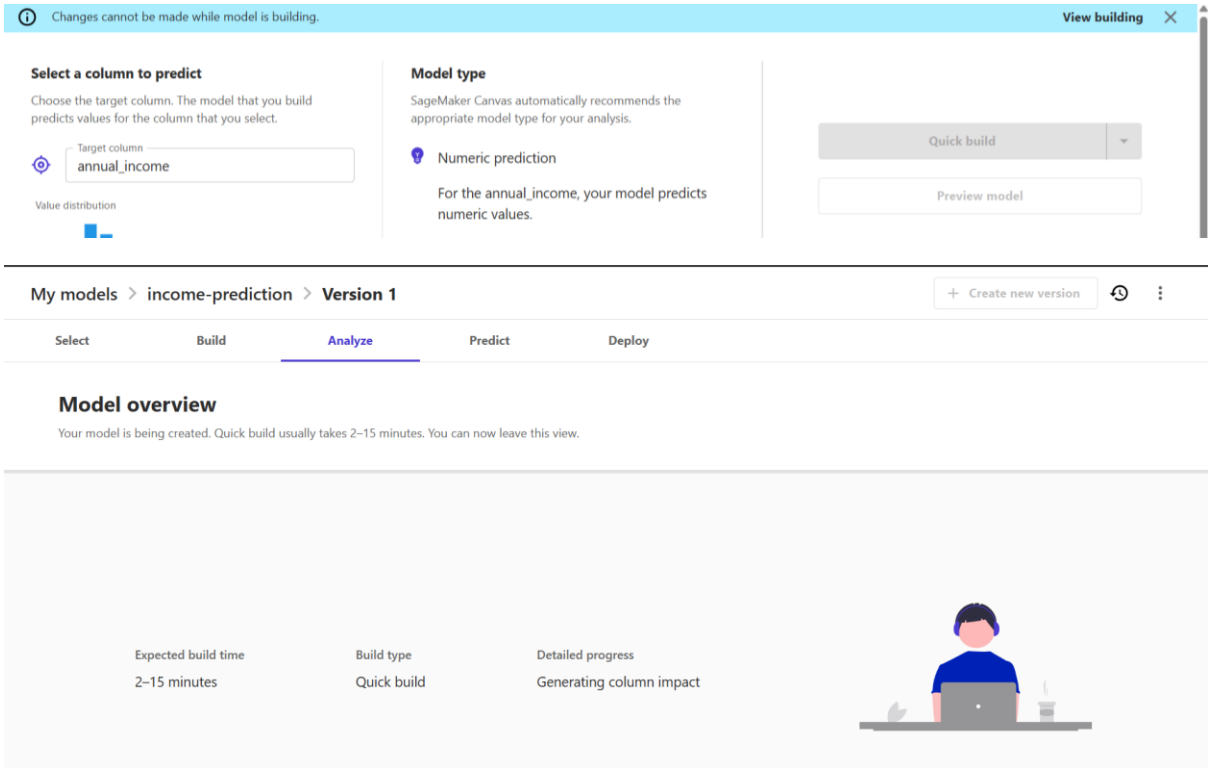


- You will also see:
- Feature/Column importance
- Data visualizer
- Explainability insights

- Use Data Visualizer to analyze data.
- Drag and drop the columns
- Bar Chart
- Compare:
- Home Ownership vs Income



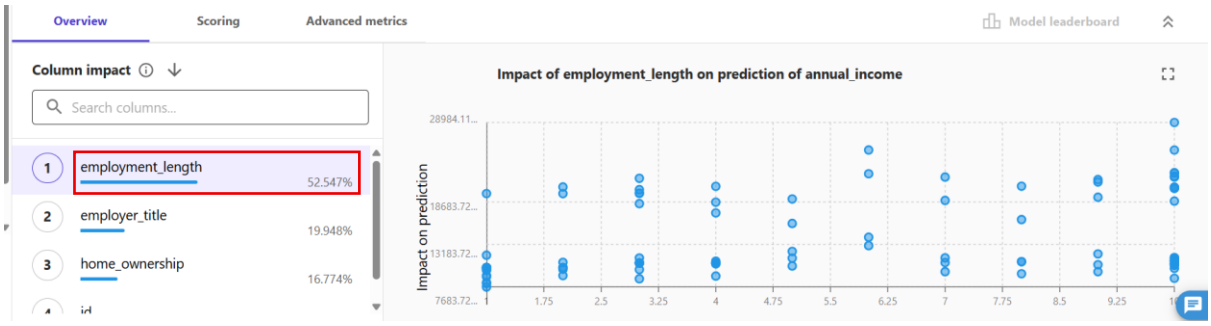
- Build the Model
- Select:
- Quick Build
- Start training.
- Typical runtime:
- 2–15 minutes



- Amazon SageMaker Canvas analyses feature importance to show which columns, such as Employment Length, Home Ownership, and Loan Amount, most influence predictions.

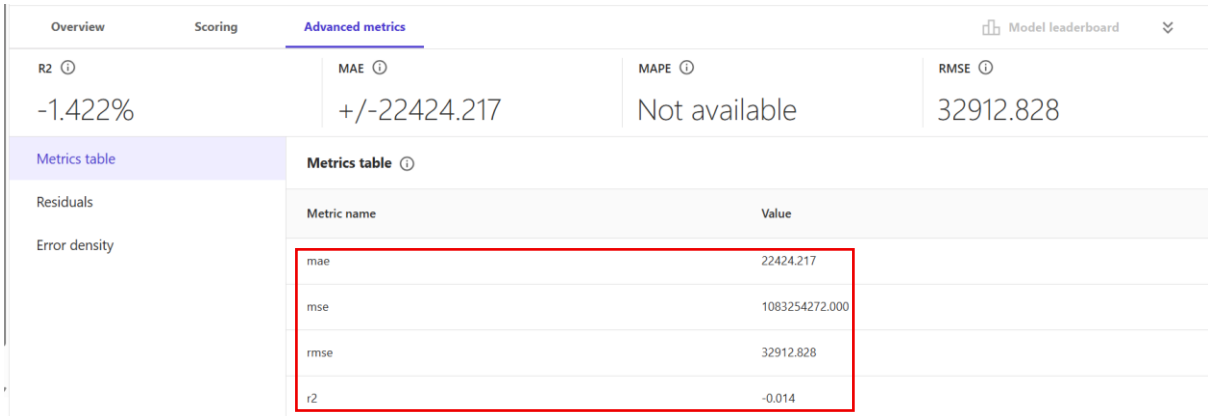
This improves model explainability, business understanding, and trust in prediction results.

So, here in this dataset employment_length has more importance.

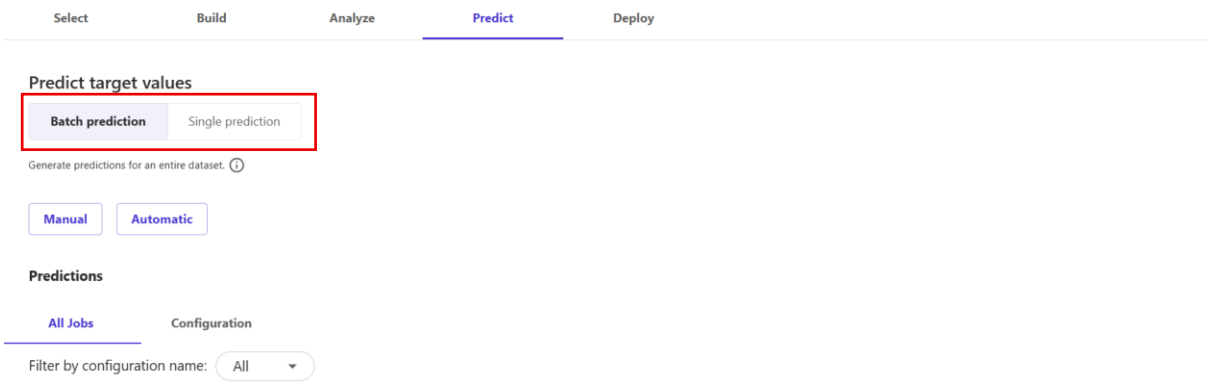


- Amazon SageMaker Canvas provides explainability visualizations such as feature impact charts, residual analysis, and predicted vs actual comparisons to evaluate model accuracy. These insights help identify correct predictions and large prediction errors, ensuring most results stay within the RMSE range.

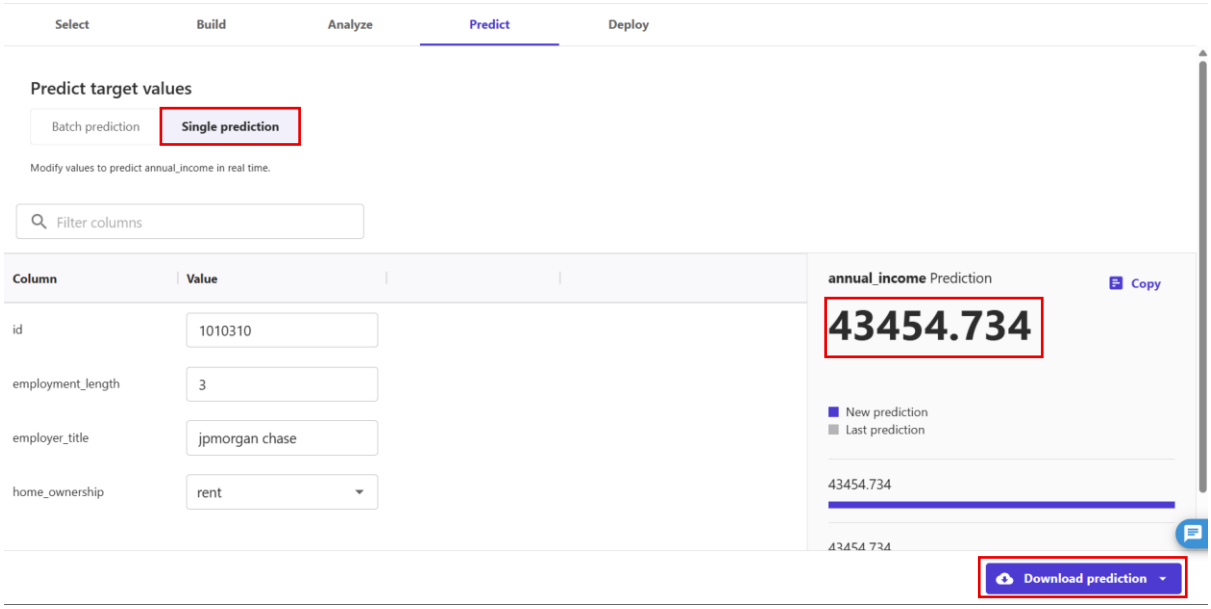
- Amazon SageMaker Canvas also offers advanced metrics such as error density, residual distributions, and additional performance statistics, providing deeper insights into overall model quality and prediction accuracy.



- Amazon SageMaker Canvas allows predictions through Batch Prediction for large datasets and automated scheduled processing, or Single Prediction for manual real-time testing. Users can either run predictions on datasets automatically or enter feature values manually to receive instant predicted income results.

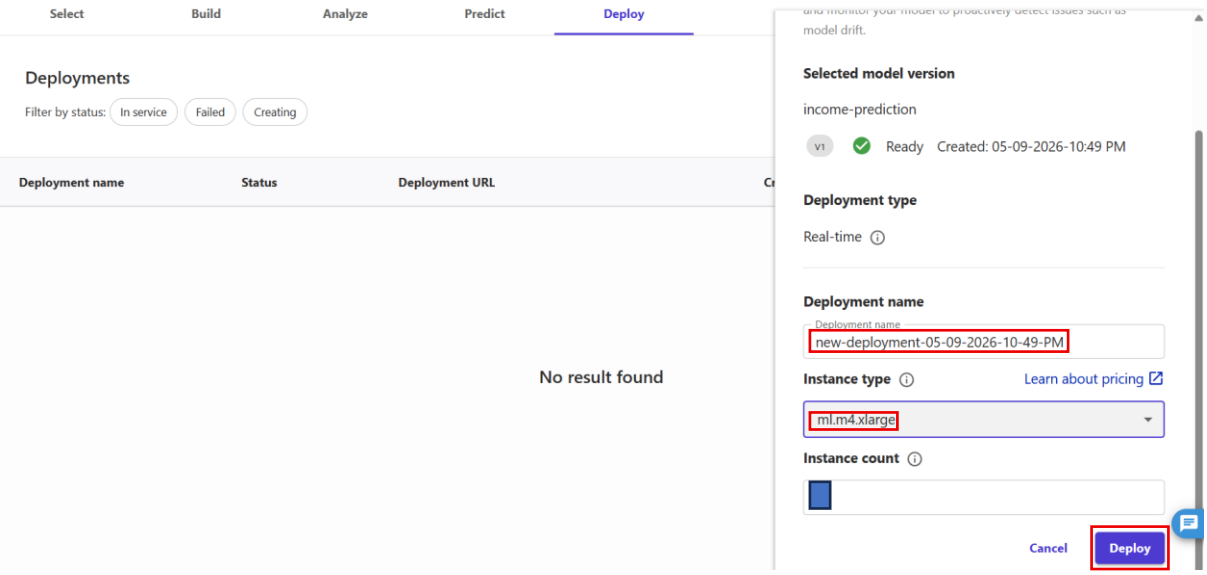
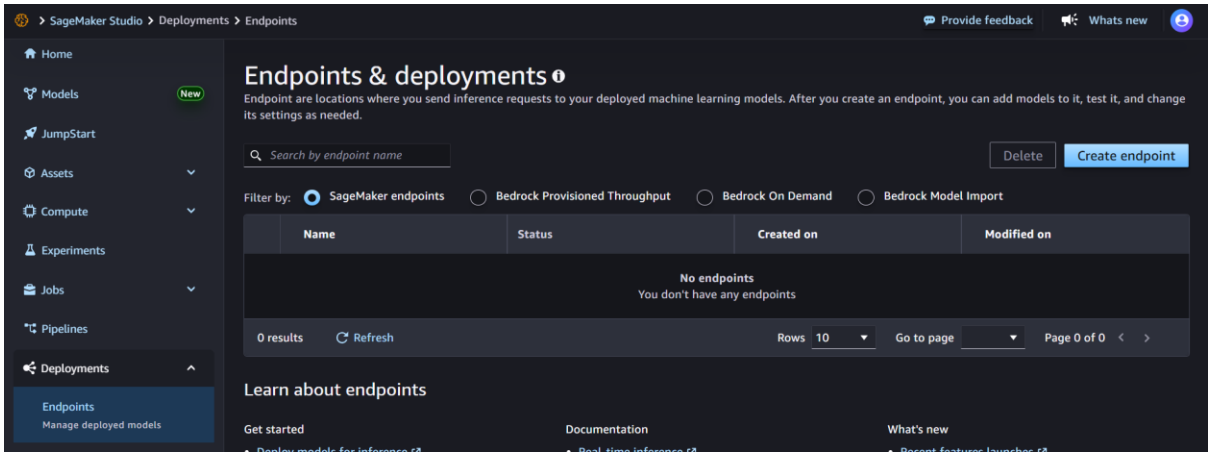


- Click on Single prediction and make the prediction also you can download it.



- Click:
- Deploy
- Click on Create Deployment.
- Canvas creates a SageMaker endpoint.

- In Amazon SageMaker Canvas, deployment configuration involves selecting the instance type and instance count based on performance and cost requirements. Smaller instances like ml.m5.large are suitable for low-cost testing, while larger or multiple instances provide better scalability and performance for production workloads.



- After deployment in Amazon SageMaker Canvas, a SageMaker endpoint is created and becomes available for inference requests. The endpoint can then be integrated with applications, accessed through APIs, and used to perform real-time predictions efficiently.